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NeoColab_REC_CS23231_DATA STRUCTURES

REC_DS using C_week 1_CY

Attempt : 1
Total Mark : 30
Marks Obtained : 11.5

Section 1 : Coding

1. Problem Statement

Timothy wants to evaluate polynomial expressions for his mathematics homework. He needs a program that allows him to input the coefficients of a polynomial based on its degree and compute the polynomial's value for a given input of x . Implement a function that takes the degree, coefficients, and the value of x , and returns the evaluated result of the polynomial.

Example

Input:

degree of the polynomial = 2

coefficient of x^2 = 13

coefficient of x^1 = 12

coefficient of $x_0 = 11$

$x = 1$

Output:

36

Explanation:

Calculate the value of $13x^2$: $13 * 12 = 13$.

Calculate the value of $12x^1$: $12 * 11 = 12$.

Calculate the value of $11x^0$: $11 * 10 = 11$.

Add the values of x^2 , x^1 , and x^0 together: $13 + 12 + 11 = 36$.

Input Format

The first line of input consists of an integer representing the degree of the polynomial.

The second line consists of an integer representing the coefficient of x^2 .

The third line consists of an integer representing the coefficient of x^1 .

The fourth line consists of an integer representing the coefficient of x^0 .

The fifth line consists of an integer representing the value of x , at which the polynomial should be evaluated.

Output Format

The output is an integer value obtained by evaluating the polynomial at the given value of x .

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 2

13

12

11

1

Output: 36

Answer

```
// You are using GCC
#include<stdio.h>
int main(){
    int d,x,i,result=0;
    scanf("%d",&d);
    int coeffs[d+1];

    for(i=d;i>=0;i--){
        scanf("%d",&coeffs[i]);
    }
    scanf("%d",&x);

    for(i=d;i>=0;i--)
        result=result*x+coeffs[i];

    printf("%d\n",result);
    return 0;
}
```

Status : Correct

Marks : 10/10

2. Problem Statement

Lisa is studying polynomials in her class. She is learning about the multiplication of polynomials.

To practice her understanding, she wants to write a program that multiplies two polynomials and displays the result. Each polynomial is represented as a linked list, where each node contains the coefficient and exponent of a term.

Example

Input:

4 3

y

3 1

y

1 0

n

2 2

y

3 1

y

2 0

n

Output:

$$8x^5 + 12x^4 + 14x^3 + 11x^2 + 9x + 2$$

Explanation

1. Poly1: $4x^3 + 3x + 1$

2. Poly2: $2x^2 + 3x + 2$

Multiplication Steps:

1. Multiply $4x^3$ by Poly2:

$$\rightarrow 4x^3 * 2x^2 = 8x^5$$

$$\rightarrow 4x^3 * 3x = 12x^4$$

$$\rightarrow 4x^3 * 2 = 8x^3$$

2. Multiply $3x$ by Poly2:

$$\rightarrow 3x * 2x^2 = 6x^3$$

$$\rightarrow 3x * 3x = 9x^2$$

$$\rightarrow 3x * 2 = 6x$$

3. Multiply 1 by Poly2:

$$\rightarrow 1 * 2x^2 = 2x^2$$

$$\rightarrow 1 * 3x = 3x$$

$$\rightarrow 1 * 2 = 2$$

Combine the results: $8x^5 + 12x^4 + (8x^3 + 6x^3) + (9x^2 + 2x^2) + (6x + 3x) + 2$

The combined polynomial is: $8x^5 + 12x^4 + 14x^3 + 11x^2 + 9x + 2$

Input Format

The input consists of two sets of polynomial terms.

Each polynomial term is represented by two integers separated by a space:

- The first integer represents the coefficient of the term.
- The second integer represents the exponent of the term.

After entering a polynomial term, the user is prompted to input a character indicating whether to continue adding more terms to the polynomial.

If the user inputs 'y' or 'Y', the program continues to accept more terms.

If the user inputs 'n' or 'N', the program moves on to the next polynomial.

Output Format

The output consists of a single line representing the resulting polynomial after multiplying the two input polynomials.

Each term of the resulting polynomial is formatted as follows:

- The coefficient and exponent are separated by 'x^' if the exponent is greater than 1.
- If the exponent is 1, only 'x' is displayed without the exponent.
- If the exponent is 0, only the coefficient is displayed.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 4 3

y

3 1

y

1 0

n

2 2

y

3 1

y

2 0

n

Output: $8x^5 + 12x^4 + 14x^3 + 11x^2 + 9x + 2$

Answer

```
#include<stdio.h>
int main(){
    int p1[11]={0},p2[11]={0},res[21]={0},c,e;
    char cont;
    do{scanf("%d %d %c",&c,&e,&cont);
    p1[e]+=c;

}while(cont=='y' || cont=='Y');

    do{scanf("%d %d %c",&c,&e,&cont);
    p2[e]+=c;

}while(cont=='y' || cont=='Y');
```

```

for (int i=0;i<=10;i++){
    for(int j=0;j<=10;j++){
        res[i+j]+=p1[i]*p2[j];
    }
}
int first=1;
for(int i=20;i>=0;i--){
    if (res[i]){
        if(!first)printf(" + ");
        first=0;
        if(i==0)
            printf("%d",res[i]);
        else if(i==1)
            printf("%dx",res[i]);
        else printf("%dx^%d",res[i],i);
    }
    printf("\n");
}

```

Status : Partially correct

Marks : 1.5/10

3. Problem Statement

John is working on a math processing application, and his task is to simplify polynomials entered by users. The polynomial is represented as a linked list, where each node contains two properties:

Coefficient of the term.

Exponent of the term.

John's goal is to combine all the terms that have the same exponent, effectively simplifying the polynomial.

Input Format

The first line of input consists of an integer representing the number of terms in the polynomial.

The next n lines of input consist of two integers, representing the coefficient and exponent of the polynomial in each line separated by space.

Output Format

The first line of output prints the original polynomial in the format 'cx^e + cx^e + ...' (where c is the coefficient and e is the exponent of each term).

The second line of output displays the simplified polynomial in the same format as the original polynomial.

If the polynomial is 0, then only '0' will be printed.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 3

5 2

3 1

6 2

Output: Original polynomial: $5x^2 + 3x^1 + 6x^2$

Simplified polynomial: $11x^2 + 3x^1$

Answer

```
#include<stdio.h>
/*#include<stdlib.h>
typedef struct Poly{
    int coeff;
    int expo;
    struct Poly *Next;
}Poly;
```

```
void create(Poly *);
void display(Poly *);
void simplify(Poly *poly1,Poly *result);
```

```
int main(){
    int n;
    Poly *poly1=(Poly*)malloc(sizeof(Poly));
    Poly *result=(Poly*)malloc(sizeof(Poly));
    poly1->Next=NULL;
    result->Next=NULL;
    if(scanf("%d",&n)!=1){
        return 1;
```



```

    }
    for(int i=0;i<n;i++){
        create(poly1);
    }
    printf("Original polynomials:");
    display(poly1);
    simplify(poly1,result);
    printf("Simplified polynomial:");
    display(result);
    return 0;
}

```

```

void display(Poly *List){
    Poly *position=List->Next;
    while(position!=NULL){
        printf("%dx^%d",position->coeff,position->expo);
        if(position->Next!=NULL && position->Next->coeff>0){
            printf("+");
        }
    }
    printf("\n");
}

```

```

void simplify(Poly *poly1,Poly *result){
    Poly *pos1=poly1->Next;
    //Poly *temp;
    //Poly *rpos=result;
    while(pos1!=NULL){
        Poly *rpos=result,
        *prev=NULL;
        while(rpos->Next!=Null && rpos->Next->expo > pos1->expo){
            prev=rpos;
            rpos=rpos->Next;
        }
        if(rpos->Next!=NULL && rpos->Next->expo == pos1->expo){
            rpos->Next->coeff+=pos1->coeff;
        }
        else{
            Poly *newnode(Poly*)malloc(sizeof(Poly));
            newnode->coeff=pos1->coeff;
            newnode->expo=pos1->expo;
            newnode->Next=rpos->Next;
        }
    }
}

```

```

    rpos->Next=newnode;
}
temp=result->Next;
while(temp!=NULL){
    if(temp->expo==pos1->expo){
        temp->coeff+=pos1->coeff;
        break;
    }
    temp=temp->Next;
}
if(temp==NULL){
    Poly *newnode=(Poly*)malloc(sizeof(Poly));
    newnode->coeff=pos1->coeff;
    newnode->expo=pos1->expo;
    newnode->Next=NULL;
    rpos->Next=newnode;
    rpos=newnode;
}
pos1=pos1->Next;
}
}

void create(Poly *List){
    Poly *position,*newnode;
    position=List;
    newnode=(Poly*)malloc(sizeof(Poly));
    if(scanf("%d %d",&newnode->coeff,&newnode->expo)!=2){
        free(newnode);
        return;
    }
    newnode->Next=NULL;
    while(position->Next!=NULL){
        position=position->Next;
    }
    position->Next=newnode;
}*/

int main(){
    int n,c,e,terms[101]={0};
    scanf("%d",&n);
    printf("Original polynomial: ");
    for(int i=0;i<n;i++){
        scanf("%d %d",&c,&e);
        terms[e]+=c;
    }
}

```

```
    if(i>0)printf(" + ");
    printf("%dx^%d",c,e);
}
printf("\nSimplified polynomial: ");
int first=1;
for(int i=100;i>=0;i--){
    if(terms[i]){
        if(!first)printf(" + ");
        first=0;
        printf("%dx^%d",terms[i],i);
    }
}
if(first)printf("0x^0");
printf("\n");}
```

Status : Wrong

Marks : 0/10